SGA 5 – Batch 028, continuation reader

Exposé VII continuation through Exposé XV

Working English LaTeX translation

This reader continues after the bridge file in this package. It translates the source range from the Exposé VII Chern-class continuation through the end of the current SGA 5 working source, including Exposés VIII, XI, XII, and XV.

Exposé VII – Chern Classes, Self-Intersection, Flag Schemes, and Blow-Ups

The source block available here resumes in the middle of the proof of the projective bundle formula, at the continuation of 2.2.2. The translation follows that source block and then continues through the end of the exposé.

2.2. Projective bundles: continuation

Let $p : P \to S$ be the projective bundle associated with a locally free $\mathcal{O}_S$-module E of rank $r + 1$, and let

$$\xi = c_1(\mathcal{O}_P(1)) \in H^2(P, \mu_n).$$

The remaining point in the proof of the projective bundle theorem is to show that, for $0 \leq i \leq r$, the class ξ^i is a free element over the cohomology ring of S. It is enough to verify this for ξ^r, since the lower powers are then handled by the usual triangular argument. This is exactly the compatibility of the trace morphism with the fundamental class in the diagram

$$\begin{array}{ccc} H^{2r}(P,\mu_n^{\otimes r}) & \xrightarrow{\mathrm{Tr}} & A \\ \cup p^*(x)\uparrow & & \uparrow x \\ H^0(P,\mathbb{Z}/n\mathbb{Z}) & \xrightarrow{\sim} & A, \end{array}$$

where p^* is induced by a closed point of the fiber.

Corollary 2.2.4. Let L be a complex of A-modules. The morphism

$$R\Gamma_*(S,L) \longrightarrow R\Gamma_*(P,p^*L)$$

induced by the canonical splitting of the projective bundle formula is an isomorphism after applying the corresponding projector.

Indeed, it suffices to apply $R\Gamma_*$ to the isomorphism furnished by the projective bundle formula.

For later use set

$$A^*(X) = \bigoplus_i H^{2i}(X,\mu_n^{\otimes i}).$$

This notation will be used systematically.

Corollary 2.2.6. The pullback homomorphism

$$p^* : A^*(S) \longrightarrow A^*(P)$$

is an injective homomorphism of unital rings, and $1,\xi,\ldots,\xi^r$ form a basis of the $A^*(S)$-module $A^*(P)$. In particular $A^*(P)$ is free over $A^*(S)$ of rank equal to the rank of E.

This is the form of the projective bundle theorem used in the construction of Chern classes and in the splitting principle below.

3. Chern classes

Let S be a scheme and let E be a locally free $\mathcal{O}_S$-module of constant rank r. Put $P = \mathbb{P}(E)$ and let $\xi = c_1(\mathcal{O}_P(1))$. The projective bundle formula shows that there is a unique relation

$$\xi^r - c_1\xi^{r-1} + c_2\xi^{r-2} - \cdots + (-1)^r c_r = 0,$$

with

$$c_i \in H^{2i}(S, \mu_n^{\otimes i}).$$

If E has non-constant rank, the scheme S is partitioned into open-and-closed subsets S_r on which E has rank r. Since cohomology decomposes over this disjoint union, the preceding construction defines the classes on each component and hence on S.

Definition 3.2. For a locally free $\mathcal{O}_S$-module E, the element $c_i(E) \in H^{2i}(S, \mu_n^{\otimes i})$ obtained by the preceding construction is called the ith Chern class of E. If the rank of E is bounded, the total Chern class is

$$c(E) = \sum_i c_i(E) \in A^*(S).$$

It is immediate from the definition that $c_0(E) = 1$, and if E has rank at most d, then $c_i(E) = 0$ for $i > d$.

Proposition 3.4. The Chern classes satisfy the following properties.

(i) Normalization: if L is invertible and $[L]$ is the image of the class of L under the boundary map

$$H^1(S, \mathcal{O}_S^*) \to H^2(S, \mu_n),$$

then $c(L) = 1 + [L]$.

(ii) Base change: for $f : S' \to S$, one has $f^*c_i(E) = c_i(f^*E)$.

(iii) Additivity: for an exact sequence of locally free modules

$$0 \to E' \to E \to E'' \to 0,$$

one has

$$c_n(E) = \sum_{i+j=n} c_i(E')c_j(E'').$$

Proof. For line bundles the projective bundle is S itself and $\mathcal{O}_P(1)$ is identified with the dual of the given line bundle, giving the normalization. Base change follows immediately from the functoriality of $\mathbb{P}(E)$, of $\mathcal{O}(1)$, and of the defining relation.

For additivity one first reduces, by partitioning S into open-and-closed pieces, to the case where all ranks are constant. Then one applies the splitting principle. Let $h : Z \to S$ be the product of the flag schemes of E' and E''. The pullbacks h^*E' and h^*E'' have finite filtrations with invertible quotients, say L_i and M_j. The pullback h^*E has a filtration with consecutive quotients the L_i and M_j. The lemma below gives

$$c(h^*E) = \prod_i (1 + [L_i]) \prod_j (1 + [M_j]) = c(h^*E')c(h^*E'').$$

The morphism h is a composite of projective-bundle morphisms, so $h^* : A^*(S) \to A^*(Z)$ is injective by Corollary 2.2.6. The desired identity follows.

Lemma 3.6. Let E be locally free and endowed with a finite filtration

$$0 = E_0 \subset E_1 \subset \cdots \subset E_r = E$$

whose successive quotients $L_i = E_i/E_{i-1}$ are invertible. Then

$$c(E) = \prod_{i=1}^{r} (1 + [L_i]).$$

The proof is the standard flag argument. On $P = \mathbb{P}(E)$ the filtration defines closed subschemes

$$S \simeq Y_1 \subset Y_2 \subset \cdots \subset Y_r = P.$$

The immersion $Y_i \hookrightarrow Y_{i+1}$ is defined by an invertible ideal identified with

$$p^*(L_i^\vee) \otimes \mathcal{O}_{Y_{i+1}}(-1).$$

Using the projection formula for the Gysin morphism and the identity $u_*(1_Z) = -[J]$ for an immersion defined by an invertible ideal J, one proves by induction the vanishing relation

$$(a_i)_* a_i^* iggl(\prod_{j \le i} p^*(L_j^\vee) \otimes \mathcal{O}_P(-1) \Bigg) = 0.$$

For $i = r$ this is precisely

$$\prod_{j=1}^{r} (\xi + c_1(p^* L_j)) = 0,$$

which is the desired expression of the defining Chern polynomial.

Corollary 3.5. For every exact sequence of locally free sheaves $0 \to E' \to E \to E'' \to 0$,

$$c(E) = c(E')c(E'')$$

in $A^*(S)$.

3.8. Comparison with classical Chern classes

If X is locally of finite type over $\operatorname{Spec} \mathbb{C}$, the analytic space $X(\mathbb{C})$ carries the usual topological theory of Chern classes. For a continuous complex vector bundle V on $X(\mathbb{C})$ one has classes

$$c_i(V) \in H^{2i}(X(\mathbb{C}), \mathbb{Z}).$$

If E is an algebraic locally free sheaf on X, its analytification gives such a bundle, and reduction modulo n gives classes in

$$H^{2i}(X(\mathbb{C}), \mathbb{Z}/n\mathbb{Z}).$$

The comparison isomorphism between analytic and étalecohomology identifies these with the étaleChern classes just defined. In particular, the square

$$\begin{array}{ccc} H^1(X, \mathcal{O}_X^*) & \longrightarrow & H^1(X(\mathbb{C}), \mathcal{O}_{X_{\mathrm{an}}}^*) \\ \partial \downarrow & & \downarrow c_1 \\ H^2(X, \mu_n) & \longrightarrow & H^2(X(\mathbb{C}), \mathbb{Z}/n\mathbb{Z}) \end{array}$$

commutes. This follows from the two Kummer-type exact sequences: algebraically,

$$0 \to \mu_n \to \mathcal{O}_X^* \xrightarrow{n} \mathcal{O}_X^* \to 0,$$

and analytically,

$$0 \to \mathbb{Z} \to \mathcal{O}_{X_{\mathrm{an}}} \xrightarrow{\exp} \mathcal{O}_{X_{\mathrm{an}}}^* \to 0.$$

Passing to the projective limit over $n = \ell^r$ defines the ℓ-adic Chern classes

$$c_i(E) \in H^{2i}(X, \mathbb{Z}_\ell(i)),$$

for every prime ℓ invertible on X.

4. The self-intersection formula and applications

Let $i : Y \hookrightarrow X$ be a regular closed immersion of codimension d between smooth schemes, and let $N = N_{Y/X}$ be the normal bundle. The cohomological Gysin morphism

$$i_* : H^m(Y, M) \to H^{m+2d}(X, M(d))$$

is characterized by the fundamental class of Y in X and by compatibility with pullback, cup-product, and trace.

Self-intersection formula. For $x \in H^*(Y, M)$ one has

$$i^* i_*(x) = c_d(N) \cup x.$$

The proof reduces by deformation to the normal bundle. In the vector-bundle case the zero-section $s : S \to N$ has normal bundle N, and the identity follows from the computation of the top Chern class through the projective completion

$$\widehat{N} = \mathbb{P}(N \oplus \mathcal{O}_S).$$

The projective bundle formula and the explicit description of the class of the zero-section give the equality.

One application is the Euler–Poincare class. If X is projective over an algebraically closed field k, the diagonal class in $X \times X$ has a self-intersection whose degree is

$$\chi_\ell(X) = \sum_i (-1)^i \dim_{\mathbb{Q}_\ell} H^i(X, \mathbb{Q}_\ell).$$

When X is projective, this class is represented in the Chow ring by the self-intersection of the diagonal cycle. Hence it is independent of the auxiliary prime $\ell \neq \mathrm{char}(k)$; one may speak simply of the Euler–Poincare characteristic $\chi(X)$.

5. Flag schemes

Let E be a locally free $\mathcal{O}_S$-module of rank n and let

$$n = p_1 + \cdots + p_k$$

be a partition. The flag scheme $D = D(p_1, \ldots, p_k; E)$ represents filtrations of E with successive quotients of ranks p_i. It is built by iterating Grassmannian or projective bundle constructions. Consequently the pullback

$$A^*(S) \longrightarrow A^*(D)$$

is injective, and $A^*(D)$ is a finite free $A^*(S)$-module with an explicit basis in Schubert monomials in the Chern classes of the tautological quotients.

The exposé follows Grothendieck's treatment of Chern classes in the Chevalley seminar. The key lemma used in flag calculations says that, for an exact sequence

$$0 \to L \to E \to F \to 0$$

with L invertible, the induced sequence of symmetric algebras

$$0 \to L \otimes S(E) \to S(E) \to S(F) \to 0$$

is exact as a sequence of graded $\mathcal{O}_S$-modules. This identifies each step in the flag tower with a regular immersion defined by an invertible ideal, allowing the Gysin formula to be applied inductively.

6. Group schemes

The same formalism applies to smooth group schemes and their homogeneous spaces. If a smooth group scheme G acts transitively on a smooth scheme X with stabilizer H, the quotient geometry gives a comparison between the cohomology of X, of G, and of H. When the quotient is built from flag varieties or from projective bundles, the projective-bundle theorem and Chern-class formulas compute its cohomology ring.

The point for later use is not merely the calculation itself but the functorial behavior: characteristic classes of the associated vector bundles are compatible with change of group, restriction to subgroups, and passage to quotient varieties. These compatibilities feed into the trace and Lefschetz formulas where equivariance is present.

7. Complete intersections

Let X be a smooth projective scheme and let $Y \subset X$ be a smooth complete intersection cut out by a regular sequence, or more generally by a regular section of a locally free sheaf. The normal bundle of Y in X is then identified with the restriction of that locally free sheaf. The self-intersection formula gives

$$i^* i_*(1) = c_d(N_{Y/X}).$$

Lefschetz theorem 7.1. For a smooth hyperplane section or suitable complete intersection, the pullback on cohomology is an isomorphism in the usual stable range and is injective at the middle degree boundary.

The exposé treats this as the étaleanalogue of the classical Lefschetz theorem. It also records the expected stronger Lefschetz statements suggested by the transcendental theory. The point is that Chern classes and Gysin morphisms supply the algebraic incarnation of the topological cycle classes used in the classical proof.

In particular, the Euler–Poincare characteristic of complete intersections is computed by applying Gauss–Bonnet to the total Chern class of the tangent bundle, with the normal exact sequence

$$0 \to T_Y \to T_X|_Y \to N_{Y/X} \to 0$$

and the multiplicativity of total Chern classes.

8. Blown-up varieties

Let X be regular and let $Y \subset X$ be a regular closed subscheme of codimension d. Let

$$p : X' \to X$$

be the blow-up of X along Y, and let $Y' = p^{-1}(Y)$ be the exceptional divisor. Then

$$Y' \simeq \mathbb{P}(N_{Y/X}),$$

where $N_{Y/X}$ is the normal bundle. Let $q : Y' \to Y$ be the projection and let $j : Y' \hookrightarrow X'$ be the exceptional divisor immersion.

The cohomology of X' is determined by that of X and Y. The additive decomposition is

$$H^*(X') \simeq p^* H^*(X) \oplus \bigoplus_{i=1}^{d-1} j_*\big(q^* H^{*-2i}(Y)(-i)\, \xi^{i-1}\big),$$

where $\xi = c_1(\mathcal{O}_{Y'}(1))$. The proof follows from the localization exact sequence for the pair (X', Y'), the corresponding sequence for (X, Y), and the projective bundle formula for $Y' \to Y$.

The multiplicative structure is determined by three rules. First,

$$p^*(a)p^*(b) = p^*(ab).$$

Second, the projection formula gives

$$p^*(a)\, j_*(z) = j_*(j^*p^*(a)\, z) = j_*(q^*i^*(a)\, z).$$

Third, products of exceptional terms are reduced to the self-intersection formula for j:

$$j^*j_*(z) = c_1(\mathcal{O}_{Y'}(-1))\, z = -\xi z.$$

Together with the Chern polynomial relation of the projective bundle $\mathbb{P}(N_{Y/X})$, these rules give the full ring structure.

9. Chow ring of a blow-up and self-intersection in the Chow ring

The last part of the exposé repeats the preceding calculation inside the Chow ring. Let X be noetherian regular, let $Y \subset X$ be regular of codimension d, let X' be the blow-up of X along Y, and let Y' be the exceptional divisor. Again $Y' \simeq \mathbb{P}(N_{Y/X})$.

Theorem 9.1. With the preceding notation, the homomorphism

$$A^*(X) \oplus \bigoplus_{i=1}^{d-1} A^{*-i}(Y) \longrightarrow A^*(X')$$

which sends $(a, b_1, \ldots, b_{d-1})$ to

$$p^*a + \sum_{i=1}^{d-1} j_*(q^*b_i\, \xi^{i-1})$$

is an isomorphism of abelian groups.

The proof is the Chow-theoretic version of the cohomological calculation. It uses the deformation to the normal cone and the projective bundle formula in Chow groups. The key computational lemma is the following.

Lemma 9.1. Let S be smooth and quasi-projective over a separably closed field, let E be locally free of rank $r+1$, and let $p : P = \mathbb{P}(E) \to S$. Let F be defined by

$$0 \to F \to p^*E \to \mathcal{O}_P(1) \to 0,$$

and put $\xi = c_1(\mathcal{O}_P(1))$. Then, for $y \in C^*(S)$,

$$y = p_!\big(p^*y\, c_r(F)\big).$$

If $E = N \oplus \mathcal{O}_S$ and $s : S \to P$ is the zero-section in the projective completion $\widehat{N}$, then

$$s_!(y) = p^*y\, \xi^r, \qquad s^*s_!(y) = y\, c_r(N^\vee),$$

and for $z \in C^*(\widehat{N})$,

$$s^*(z) = p_!\big(z\, c_r(F)\big).$$

This lemma is a direct analogue of the cohomological computation in the normal bundle. It reduces the self-intersection formula to the projective bundle calculation and the behavior of the zero-section.

The self-intersection formula in the Chow ring. Let $i : Y \hookrightarrow X$ be a regular closed immersion of codimension d, with normal bundle N. For $y \in A^*(Y)$ one has

$$i^*i_*(y) = c_d(N)\, y$$

in $A^*(Y)$.

This is Mumford's Chow-theoretic self-intersection formula. It is obtained by applying the preceding blow-up computation to the deformation to the normal cone, where the assertion becomes the explicit calculation for the zero-section.

9.3. Useful relations

Inside the Chow ring of the blow-up, the exceptional divisor class and the Chern classes of the normal bundle satisfy the relation

$$\xi^d - c_1(N)\xi^{d-1} + \cdots + (-1)^d c_d(N) = 0$$

on $Y' = \mathbb{P}(N)$. The exceptional divisor satisfies

$$j^* j_*(1) = -\xi.$$

Together with the projection formula, these relations determine all products involving exceptional classes.

9.4. Key lemma

The key lemma asserts that the kernel and cokernel appearing in the comparison of $A^*(X')$ with the direct sum of $A^*(X)$ and the shifted $A^*(Y)$ are killed successively by projective bundle trace maps. More concretely, for the filtration by powers of the exceptional divisor, the associated graded pieces are the corresponding graded pieces of

$$A^*(Y')[1, \xi, \ldots, \xi^{d-1}],$$

modulo the Chern polynomial. The projective bundle formula identifies these pieces with $A^*(Y)$.

9.5. End of the proof

The proof of the blow-up theorem follows by combining the key lemma with the localization exact sequences

$$A^*(Y) \to A^*(X) \to A^*(X \setminus Y) \to 0$$

and

$$A^*(Y') \to A^*(X') \to A^*(X' \setminus Y') \to 0,$$

noting that $X' \setminus Y'$ is canonically isomorphic to $X \setminus Y$. The resulting diagram has exact rows. The projective bundle formula computes the left vertical term, and the five lemma gives the decomposition.

9.6. The key formula

The fundamental identity may be written in the compact form

$$j_*\big(q^*(y)\xi^m\big) \cdot j_*\big(q^*(z)\xi^n\big) = -j_*\big(q^*(yz)\xi^{m+n+1}\big),$$

with the understanding that powers ξ^a for $a \geq d$ are reduced by the Chern polynomial of N. This formula is simply the projection formula and the equality $j^*j_*(1) = -\xi$.

9.8. Proof of Proposition 9.6

Proposition 9.6 is proved by applying the preceding formula repeatedly and using the fact that $q^* : A^*(Y) \to A^*(Y')$ is injective. The exceptional contribution is expressed in the basis $1, \xi, \ldots, \xi^{d-1}$ over $A^*(Y)$; uniqueness of the expression follows from the projective bundle theorem. The compatibility with pushforward to X comes from

$$p_*j_* = i_*q_*.$$

9.9. Structure theorem for the Chow group of the blow-up

The structure theorem states that $A^*(X')$ is the direct sum of the pullback of $A^*(X)$ and the exceptional summands generated by $j_*(q^*A^*(Y)\xi^i)$ for $0 \leq i \leq d-2$. Multiplication is described by the formulas above. Equivalently, $A^*(X')$ is the $A^*(X)$-algebra generated by the exceptional divisor class, with relations induced by the Chern polynomial of N and the restriction of cycles from X to Y.

9.10. Interpretation and Riemann–Roch

The final interpretation identifies the blow-up formula with the Riemann–Roch behavior of the exceptional divisor and the normal bundle. The Chern character and Todd class transform as dictated by the exact sequences

$$0 \to T_{X'} \to p^*T_X \to \text{exceptional quotient} \to 0$$

and

$$0 \to T_Y \to T_X|_Y \to N \to 0.$$

Thus the blow-up formula is compatible with the expected Riemann–Roch identity: pushforward of characteristic classes through p recovers the classes on X, while the exceptional summands account for the correction terms supported on Y.

Bibliography of Exposé VII

The sources cited include Grothendieck's theory of Chern classes, the Chevalley seminar on Chow rings and applications, Mumford's proof of the self-intersection formula in the Chow ring, and the Lefschetz theorems used in the cohomological applications.

Exposé VIII – Class Groups of Abelian and Triangulated Categories. Perfect Complexes

Class groups of abelian and triangulated categories. Perfect complexes.

by A. Grothendieck, written by I. Bucur

This exposé recalls, in a rapid form, the notions and results needed later concerning Grothendieck groups, pseudo-coherent complexes, and perfect complexes. A more complete treatment, with detailed proofs, is announced for the first exposés of the seminar on the Riemann–Roch theorem for schemes, later SGA 6.

1. Abelian categories

Let $\mathcal{C}$ be an abelian category and $\mathcal{D}$ a full subcategory of $\mathcal{C}$. A function f from $\mathcal{D}$ to an abelian group G is called *additive* if, for every exact sequence

$$0 \to E' \to E \to E'' \to 0$$

in $\mathcal{C}$ whose three terms belong to $\mathcal{D}$, one has

$$f(E) = f(E') + f(E'').$$

From the definition one immediately obtains $f(0) = 0$, invariance under isomorphism, and the corresponding formula for exact sequences viewed in the opposite category. These elementary properties are the universal motivation for the Grothendieck group.

2. The Grothendieck group of an abelian category

Let $\mathcal{C}$ be an abelian category. Let $F(\mathcal{C})$ be the free abelian group on isomorphism classes $[E]$ of objects of $\mathcal{C}$. The Grothendieck group $K(\mathcal{C})$ is the quotient of $F(\mathcal{C})$ by the subgroup generated by the elements

$$[E] - [E'] - [E'']$$

for all exact sequences $0 \to E' \to E \to E'' \to 0$. The canonical map

$$k_{\mathcal{C}} : \mathrm{Ob}\,\mathcal{C}/\simeq \longrightarrow K(\mathcal{C})$$

is universal for additive functions out of $\mathcal{C}$: if f is additive with values in an abelian group G, then there is a unique homomorphism $g : K(\mathcal{C}) \to G$ such that $f = g \circ k_{\mathcal{C}}$.

If $u : \mathcal{C} \to \mathcal{C}'$ is exact, then u induces a homomorphism

$$K(u) : K(\mathcal{C}) \to K(\mathcal{C}'),$$

characterized by

$$K(u)([E]) = [u(E)].$$

This is the functoriality of the Grothendieck group in the abelian setting.

3. Triangulated categories

Let $\mathcal{C}$ be a triangulated category. The Grothendieck group $K(\mathcal{C})$ is defined as the free abelian group on isomorphism classes of objects, modulo the relations

$$[X] - [Y] + [Z] = 0$$

for every distinguished triangle

$$X \to Y \to Z \to X[1].$$

Thus $[Y] = [X]+[Z]$. An exact functor of triangulated categories $u : \mathcal{C} \to \mathcal{C}'$ induces

$$K(u) : K(\mathcal{C}) \to K(\mathcal{C}')$$

by the same rule $[E] \mapsto [u(E)]$.

If $\mathcal{C} = \mathrm{D}^b(\mathcal{A})$ is the bounded derived category of an abelian category, the canonical functor from $\mathcal{A}$ to $\mathrm{D}^b(\mathcal{A})$ induces an isomorphism

$$K(\mathcal{A}) \simeq K(\mathrm{D}^b(\mathcal{A})).$$

The class of a bounded complex $K^\bullet$ is

$$[K^\bullet] = \sum_i (-1)^i [H^i(K^\bullet)],$$

or equivalently the alternating sum of its terms whenever this makes sense.

4. Thick subcategories and localization

A full subcategory $\mathcal{C}'$ of a triangulated category $\mathcal{C}$ is *thick* if it is triangulated and closed under direct summands: any direct factor in $\mathcal{C}$ of an object of $\mathcal{C}'$ again belongs to $\mathcal{C}'$. If $\mathcal{C}'$ is thick, the quotient category

$$\mathcal{C}/\mathcal{C}'$$

is triangulated and comes with an exact functor $Q : \mathcal{C} \to \mathcal{C}/\mathcal{C}'$ which is universal among exact functors from $\mathcal{C}$ annihilating $\mathcal{C}'$.

There is a natural exact sequence

$$K(\mathcal{C}') \longrightarrow K(\mathcal{C}) \xrightarrow{K(Q)} K(\mathcal{C}/\mathcal{C}') \longrightarrow 0.$$

This is the categorical form of localization. It will later be used to pass from complexes with torsion, or with prescribed support, to quotient categories where those complexes vanish.

5. Pseudo-coherent complexes

Let $\mathcal{C}$ be an abelian category and let $\mathcal{C}_0$ be a full subcategory satisfying stability conditions under kernels, cokernels, and the subquotients needed to perform homological constructions. We denote by

$$\mathrm{D}^{\pm}_{\mathcal{C}_0}(\mathcal{C})$$

the essential image of $\mathrm{D}^{\pm}(\mathcal{C}_0)$ in $\mathrm{D}(\mathcal{C})$. Its objects are called $\mathcal{C}_0$-pseudo-coherent complexes. Equivalently, a complex of $\mathcal{C}$ is $\mathcal{C}_0$-pseudo-coherent if it is isomorphic in $\mathrm{D}(\mathcal{C})$ to a complex bounded above whose terms lie in $\mathcal{C}_0$.

This subcategory is triangulated. Indeed, if two terms in a distinguished triangle

$$U \to V \to W \to U[1]$$

are pseudo-coherent, represent them by complexes $X^\bullet$ and $Y^\bullet$ with terms in $\mathcal{C}_0$. After replacing $X^\bullet$ by a quasi-isomorphic complex, the morphism is represented by an actual morphism of complexes. Its mapping cone has terms in $\mathcal{C}_0$ by the stability assumptions, and represents the third object W.

The intersection of $\mathrm{D}^{\pm}_{\mathcal{C}_0}(\mathcal{C})$ with $\mathrm{D}^b(\mathcal{C})$ is denoted

$$\mathrm{D}^{\pm,b}_{\mathcal{C}_0}(\mathcal{C}).$$

An object M of $\mathcal{C}$ is pseudo-coherent if it is pseudo-coherent as a complex concentrated in degree zero; equivalently, it admits a resolution on the left by objects of $\mathcal{C}_0$.

Proposition 5.1. A complex $X^\bullet$ is $\mathcal{C}_0$-pseudo-coherent if and only if it is isomorphic to a bounded-above complex whose terms are $\mathcal{C}_0$-pseudo-coherent objects.

The proof is deferred to SGA 6, where the theory is treated systematically.

6. Perfect complexes

An object of $\mathrm{D}(\mathcal{C})$ is called $\mathcal{C}_0$-perfect if it is isomorphic to an object of $\mathrm{D}^b(\mathcal{C}_0)$, that is, to a bounded complex all of whose terms lie in $\mathcal{C}_0$. The full subcategory of perfect complexes is denoted

$$\mathrm{D}^{\mathrm{parf}}_{\mathcal{C}_0}(\mathcal{C}).$$

It is the essential image of $\mathrm{D}^b(\mathcal{C}_0)$ in $\mathrm{D}(\mathcal{C})$. It is a triangulated subcategory, stable under direct summands under the standard hypotheses. The Grothendieck group of perfect complexes is therefore defined by the preceding triangulated construction.

When $\mathcal{C}$ is a category of modules and $\mathcal{C}_0$ is the category of finite free or finite projective modules, this recovers the usual perfect complexes. In geometry, perfect complexes are precisely the complexes locally quasi-isomorphic to bounded complexes of finite locally free modules.

7. An important special case

Let A be a ring and let $\mathcal{C}$ be the category of A-modules. Let $\mathcal{C}_0$ be the category of finite projective A-modules. Then a complex is perfect if and only if it is locally, for the relevant topology, quasi-isomorphic to a bounded complex of finite projective modules. The resulting Grothendieck group is the K-group used in Riemann–Roch formulas.

This special case is central because cohomology complexes with constructible coefficients often become perfect after imposing finite Tor-dimension conditions. In that situation, the trace of an endomorphism is computed in the Grothendieck group and then evaluated by a rank, character, or local term.

8. Tor of complexes

Let $\mathcal{C}_\ell$ denote a category of modules over a coefficient ring such as $\mathbb{Z}_\ell$ or $\mathbb{Z}/\ell^n\mathbb{Z}$. A complex X of $\mathrm{D}(\mathcal{C}_\ell)$ has *finite Tor-dimension* if it is bounded and if there exists an integer n_0 such that

$$\mathrm{Tor}^i(X, M) = 0$$

for every object M and every $i > n_0$ after the usual shift convention. Equivalently, tensoring with X has uniformly bounded cohomological amplitude.

Finite Tor-dimension is the condition which permits one to define derived tensor products without leaving bounded categories. It is also the condition which makes duality behave as expected: a perfect complex has finite Tor-dimension, and its dual is again perfect.

9. Reminders on linear representations of finite groups

Let G be a finite group and A a commutative ring. The category of finite projective $A[G]$-modules has a Grothendieck group $K(A[G])$. If A is a field of characteristic not dividing $|G|$, this group is controlled by characters. In the integral ℓ-adic setting one uses projective $\mathbb{Z}_\ell[G]$-modules and compares them after extending scalars to $\mathbb{Q}_\ell$.

Induction and restriction are exact functors and therefore induce maps on Grothendieck groups:

$$\mathrm{Ind}_H^G : K(A[H]) \to K(A[G]), \qquad \mathrm{Res}_H^G : K(A[G]) \to K(A[H]).$$

Duality sends a module M to

$$M^* = \mathrm{Hom}_A(M, A),$$

with the contragredient action. If the character of M is χ, then that of M^* is $g \mapsto \chi(g^{-1})$.

4. The Swan representation

The next part of the source turns to the local representation attached to a ramified Galois cover of curves. Let C be a smooth, proper, connected curve over an algebraically closed field k, with generic point η and function field $K = k(\eta)$. Let

$$K \subset K', \qquad [K' : K] = n$$

be a finite Galois extension with group G, and let C' be the normalization of C in K'. The canonical map is

$$p : C' \to C.$$

The group G acts on C'. For $y' \in C'$ write $G_{y'}$ for its stabilizer.

If Π is a uniformizer at y' and $s \in G_{y'}$, $s \neq e$, define

$$v_{y'}(s) = v_{y'}(s(\Pi) - \Pi).$$

This is the multiplicity of y' as a fixed point of the automorphism $s : C' \to C'$. The equality follows from the local intersection calculation: the diagonal and the graph of s meet with multiplicity $v_{y'}(s(\Pi) - \Pi)$.

4.2. Definition of the Swan representation

The Swan character attached to y' is the class function

$$\mathrm{sw}_{y'}(s) = \begin{cases} -1 - v_{y'}(s), & s \neq e, \\ \sum_{s\neq e} v_{y'}(s), & s = e. \end{cases}$$

The theorem of Artin–Serre–Swan asserts that, for a prime $\ell \neq \mathrm{char}(k)$, there is a finite projective $\mathbb{Z}_\ell[G_{y'}]$-module $\mathrm{Sw}_{y'}$, unique up to isomorphism, having this character.

One also considers

$$\widetilde{\mathrm{Sw}}_{y'} = \mathrm{Sw}_{y'} \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell.$$

Its character is

$$\widetilde{\mathrm{sw}}_{y'}(s) = \begin{cases} -1, & s \neq e, \\ |G_{y'}| - 1, & s = e. \end{cases}$$

Finally, the Artin character is

$$\mathrm{art}_{y'} = \mathrm{sw}_{y'} + \mathbf{1}_{G_{y'}},$$

where $\mathbf{1}_{G_{y'}}$ denotes the character of the augmentation representation. Explicitly,

$$\mathrm{art}_{y'}(s) = \begin{cases} -v_{y'}(s), & s \neq e, \\ \sum_{s\neq e} v_{y'}(s) + |G_{y'}|, & s = e. \end{cases}$$

Let $y = p(y')$. By induction from $G_{y'}$ to G set

$$\begin{aligned} \mathrm{Sw}_y &= \mathrm{Sw}_{y'} \otimes_{\mathbb{Z}_\ell[G_{y'}]} \mathbb{Z}_\ell[G], \\ \widetilde{\mathrm{Sw}}_y &= \widetilde{\mathrm{Sw}}_{y'} \otimes_{\mathbb{Z}_\ell[G_{y'}]} \mathbb{Z}_\ell[G], \\ \mathrm{Art}_y &= \mathrm{Art}_{y'} \otimes_{\mathbb{Z}_\ell[G_{y'}]} \mathbb{Z}_\ell[G]. \end{aligned}$$

Their characters are denoted sw_y, $\widetilde{\mathrm{sw}}_y$, and art_y.

4.3. Independence of the point above y

The induced modules just defined depend, up to isomorphism, only on $y \in C$, not on the chosen point y' above y. If $y'' = gy'$, then the situation at y'' is obtained from that at y' by the automorphism g of C' and by conjugation $s \mapsto gsg^{-1}$ on G. Extension of operators through this inner automorphism sends the corresponding module to an isomorphic one. Hence Sw_y, $\widetilde{\mathrm{Sw}}_y$, and Art_y are well-defined.

Proposition 4.4. With the preceding notation, and for $M^\bullet \in \mathrm{D}^*(\mathbb{Z}_\ell[G])$, one has canonical isomorphisms

$$\mathrm{Sw}_{y'}^* = \mathrm{Sw}_{y'},$$

and

$$\mathrm{Hom}_{\mathbb{Z}_\ell[G]}(\mathrm{Sw}_{y'}, M^\bullet) \simeq \mathrm{Sw}_{y'} \otimes_{\mathbb{Z}_\ell[G]} M^\bullet.$$

Proof. Finite projective $\mathbb{Z}_\ell[G]$-modules are determined by their characters in this setting. If σ is the character of $\mathrm{Sw}_{y'}$, then the dual has character $\sigma^*(s) = \sigma(s^{-1})$. But

$$v_{y'}(s^{-1}(\Pi) - \Pi) = v_{y'}(s^{-1}(s(\Pi) - \Pi)) = v_{y'}(s(\Pi) - \Pi),$$

so the Swan character is invariant under inversion. This gives self-duality; the Hom–tensor formula follows.

4.5. Explicit character formulas

For later use the character of the induced Swan representation can be written explicitly. For $s \in G$,

$$\mathrm{sw}_y(s) = \sum_{p(y')=y} \mathrm{sw}_{y'}(s),$$

with terms understood to be zero unless s fixes the corresponding point. Thus

$$\mathrm{sw}_y(s) = \begin{cases} \sum_{p(y')=y}(1 - v_{y'}(s)), & s \neq e, \\ 0, & s = e, \end{cases}$$

under the convention of the source. Similarly

$$\mathrm{art}_y(s) = \begin{cases} \sum_{p(y')=y}(-v_{y'}(s)), & s \neq e, \\ 0, & s = e, \end{cases}$$

and

$$\widetilde{\mathrm{sw}}_y(s) = \begin{cases} \sum_{p(y')=y}(-1), & s \neq e, \\ 0, & s = e. \end{cases}$$

These formulas are obtained from the character formula for induced representations, choosing representatives for the right cosets of $G_{y'}$ in G.

4.6. The non-connected case

The invariants Sw_y, $\widetilde{\mathrm{Sw}}_y$, and Art_y are defined under slightly more general hypotheses. Let C' be a smooth algebraic curve over k, not necessarily connected, with an action of a finite group G. Suppose there is a dense open subset U of $C = C'/G$ such that $C'|_U$ is a principal G-cover of U.

For a point $y \in C$, choose $y' \in C'$ above y. Let $C'_{y'}$ be the irreducible component of C' containing y', and C_y the component of C containing y. Let G'_y be the stabilizer of the component $C'_{y'}$. Then the extension of function fields

$$k(\eta_y) \subset k(\eta'_y)$$

is Galois with group G'_y, and $C'_{y'}$ is the normalization of C_y in that extension. Applying the connected construction to $C'_{y'} \to C_y$ gives projective $\mathbb{Z}_\ell[G'_y]$-modules. The desired $\mathbb{Z}_\ell[G]$-modules Sw_y, $\widetilde{\mathrm{Sw}}_y$, and Art_y are their inductions from G'_y to G.

Remark 4.6. The modules Sw_y, $\widetilde{\mathrm{Sw}}_y$, and Art_y are purely local invariants. They are determined by the action of the stabilizer on the completed local ring $\widehat{\mathcal{O}}_{C',y'}$. Moreover $\mathrm{Sw}_y = 0$ if and only if the cover is tamely ramified at y'; equivalently, the stabilizer $G_{y'}$ has order prime to $\mathrm{char}(k)$.

Exposé XI (end) – Euler–Poincare Characteristic and Weil Formulas

5. The Weil formula

This final part of Exposé XI compares the trace formalism with the classical formula of Weil. The situation is that of a smooth proper curve C over an algebraically closed field, an open dense subset $U \subset C$, a finite Galois covering $p : C' \to C$ with group G, and a coefficient ring Λ such as $\mathbb{Z}_\ell$ or a finite quotient. The sheaves under consideration are constructible on C, locally constant on U, and controlled near the boundary by the local Galois data of the covering.

The Euler–Poincare characteristic is written

$$\chi(C, F) = \sum_i (-1)^i [H^i(C, F)]$$

in the Grothendieck group of Λ-modules, or in the corresponding Grothendieck group of $\Lambda[G]$-modules when an equivariant structure is present. The goal is to express the difference between the global Euler characteristic and the rank contribution in terms of local ramification classes.

Proposition 5.1. Let F be a constructible sheaf on C, locally constant on U, and suppose that its pullback to U' is represented by a finite projective $\Lambda[G]$-module. Then the equivariant Euler characteristic of F is the sum of the unramified global term and local correction terms at the points of $C \setminus U$.

The proof is a formal consequence of excision and induction. One chooses a finite Galois cover which trivializes the sheaf on U, writes the cohomology on U in terms of the cohomology upstairs together with the G-module of fibers, and compares compactly supported cohomology with cohomology on the compact curve. The boundary contributions are precisely the Swan and Artin representations defined in Exposé VIII.

6. Definition of the local terms $\varepsilon_x^{\Delta}(F)$

Let $x \in C \setminus U$. The local term $\varepsilon_x^{\Delta}(F)$ is an element of a Grothendieck group, attached to the action of the decomposition group at a point x' above x. It measures the defect between the representation carried by the generic fiber of F and the representation seen after extending across x. In concrete terms it is built from the Swan representation and the fiber $F_{\bar{\eta}}$:

$$\varepsilon_x^{\Delta}(F) = [\mathrm{Sw}_x \otimes F_{\bar{\eta}}] + \text{tame correction} - \text{invariant correction}.$$

The precise expression depends on whether one writes the formula in $K(\Lambda)$, in $K(\Lambda[G])$, or after applying a character. The important point is that the local term is determined by the completed local ring and the local monodromy representation.

The notation Δ emphasizes that this term is the local contribution at the diagonal in the Lefschetz trace formula: it is the same obstruction which appears when one computes the intersection of the graph of an endomorphism with the diagonal.

7. Euler–Poincare formula

Theorem 7.1. Let C be a smooth proper connected curve over an algebraically closed field, let $U \subset C$ be a dense open subset, and let F be a constructible Λ-sheaf on C, locally constant on U. Then

$$\chi(C, F) = \mathrm{rk}(F)\chi(C, \Lambda) - \sum_{x \in C \setminus U} a_x(F),$$

where $a_x(F)$ is the Artin conductor, expressed in K-theoretic form by the local term $\varepsilon_x^{\Delta}(F)$.

Proof. The proof proceeds in four steps.

First, for a bounded-below complex Ψ of sheaves of $\Lambda[G]$-modules on C, one uses the projection formula and the exactness of induction to identify the equivariant Euler characteristic of $p_*\Psi$ with the induced class from the cohomology of Ψ upstairs.

Second, if Φ is a bounded-below complex on C which is zero outside U, its compactly supported cohomology is computed by extension by zero and the preceding equivariant comparison.

Third, the local boundary triangle

$$j_!j^*F \to F \to i_*i^*F \to$$

expresses $\chi(C, F)$ as a sum of the compactly supported contribution over U and the stalk contributions at $C \setminus U$.

Finally, by choosing a cover $U' \to U$ which trivializes the sheaf and extending to a finite cover $C' \to C$, the local terms are identified with the Swan and Artin modules. This yields the stated conductor formula.

Corollary 7.9. If F is locally constant on all of C, then all local conductor terms vanish and

$$\chi(C, F) = \mathrm{rk}(F)\chi(C, \Lambda).$$

Corollary 7.12. For a constructible sheaf F on a smooth curve, the Euler characteristic depends only on the generic rank, the Euler characteristic of the curve, and the local ramification invariants at the finitely many points where F is not locally constant.

Lemma 7.14. The local conductor class is additive in distinguished triangles and in short exact sequences of constructible sheaves. It is invariant under replacing a local trivializing cover by a dominating cover.

The proof is a diagram chase in the Grothendieck group, using exactness of restriction, induction, and the additivity of the Swan representation. This is the form which makes the Euler–Poincare formula independent of auxiliary choices.

References for Exposé XI

The references are to the preceding theory of local representations, to the Grothendieck–Ogg–Shafarevich formula, and to the trace formula for sheaves on curves.

Exposé XII – Nielsen–Wecken and Lefschetz Formulas in Algebraic Geometry

Nielsen–Wecken and Lefschetz formulas in algebraic geometry

1. General setting

The exposé studies fixed point formulas for correspondences and endomorphisms in algebraic geometry, with particular attention to local terms. The input consists of a scheme X, an endomorphism or correspondence f, a constructible sheaf or complex F, and sometimes a finite group G acting on a covering $X' \to X$. One wants a formula expressing the trace of the induced endomorphism on cohomology as a sum over fixed point classes.

In topology this is the Nielsen–Wecken refinement of the Lefschetz fixed point formula. In algebraic geometry, the role of a fixed point class is played by a connected component of the intersection of the graph of the correspondence with the diagonal, or by its lift to a finite Galois covering.

2. Fixed points and lifted fixed points

Let $f : X \to X$ be an endomorphism and let $p : X' \to X$ be a finite covering with group G. A lift $f' : X' \to X'$ need not commute with the G-action; it may instead satisfy

$$f'g = \varphi(g)f'$$

for an endomorphism $\varphi : G \to G$. For $g \in G$ one considers the fixed points of

$$g^{-1}f' : X' \to X'.$$

The local terms attached to these fixed points are then assembled into a class function on G, or into an element of a Grothendieck group after applying traces.

This is the algebraic form of separating fixed points into Nielsen classes. The summation over $g \in G$ recovers the ordinary Lefschetz number downstairs.

3. The local Nielsen–Wecken invariant

Let x be a fixed point downstairs, and choose points x' above it. The local invariant depends on the intersection multiplicity of the graph of $g^{-1}f'$ with the diagonal at x', and on the trace of the induced endomorphism on the stalk of the sheaf. In the case of curves, if R is the completed local ring, π a uniformizer, and h the induced endomorphism, the fixed point multiplicity is

$$\nu(h) = v(h(\pi) - \pi).$$

This is the same local intersection calculation used earlier for the Swan representation.

4. Generalities on local terms

For a prime $\ell \neq \mathrm{char}(k)$, the local term associated with x is denoted

$$S_{x,\ell}(f', f, G, \varphi).$$

It is a rational expression obtained by summing the local traces over the lifted fixed points.

Conjecture 4.5. For every $\ell \neq \mathrm{char}(k)$, the values of $S_{x,\ell}(f', f, G, \varphi)$ are ℓ-adic integers.

The exposé proves this conjecture in dimension one by reducing the assertion to an explicit calculation in complete discrete valuation rings.

Proposition 4.6. Let R be a discrete valuation ring with maximal ideal $\mathfrak{m}$, uniformizer π, algebraically closed residue field k, and valuation v. Let $f'_R : R \to R$ be an endomorphism and let G be a finite group of automorphisms of R. Assume that f'_R and all elements of G induce the identity on k, and that an endomorphism $\varphi : G \to G$ is given with the compatibility

$$f'_R \circ g = \varphi(g) \circ f'_R.$$

Then the local expression defining the Nielsen–Wecken term is an ℓ-adic integer.

The proof reduces to the case $g = e$ and $\varphi = \mathrm{id}$, by factoring through the kernel on which the action is relevant. Let

$$r = \nu(f'_R) = v(f'_R(\pi) - \pi).$$

If $r \geq 2$, the following lemma controls the valuations of the iterates and of the terms obtained after applying elements of G.

Lemma 4.7. Under the hypotheses of Proposition 4.6, assume in addition that $\varphi = \mathrm{id}_G$ and $r \geq 2$. Then the valuation r satisfies the stability conditions required to compute the local fixed point multiplicities, and the resulting local term is integral.

The verification is a direct calculation in R. One writes $f'_R(\pi) = \pi + u\pi^r + \cdots$ with u a unit and tracks the effect of automorphisms $g \in G$ on the leading term. The compatibility with f'_R forces the same leading valuation for the relevant differences. Thus the denominator which appears in the formal local term is cancelled by the multiplicity.

Corollary 4.8. The rational local numbers obtained from Proposition 4.6 have ℓ-adic integral images for every $\ell \neq \mathrm{char}(k)$.

Proposition 4.9. Conjecture 4.5 is true when $\dim X = 1$.

The proof introduces the endomorphism on the completed local rings at the fixed points, reduces to the complete DVR calculation, and sums over the finite set of lifted fixed points.

5. Generalization of the Nielsen–Wecken formula

Theorem 5.1. In the preceding equivariant situation, the trace of the induced endomorphism on the cohomology of X is equal to the sum of the local Nielsen–Wecken terms over the fixed point classes:

$$\mathrm{Tr}(f^* \mid R\Gamma(X, F)) = \sum_x S_{x,\ell}(f', f, G, \varphi; F).$$

Proof. One first reduces modulo ℓ^n for every $n \geq 1$, where the finite coefficient Lefschetz formula is available. The equivariant decomposition of the covering separates the trace into contributions indexed by $g \in G$. Passing to the inverse limit and using the integrality proved in dimension one gives the asserted ℓ-adic formula. Compatibility of the local terms with reduction modulo ℓ^n ensures that the resulting expression is independent of n.

5.10. Special cases

If the covering is unramified near the relevant fixed points, the Swan terms vanish and the local formula reduces to the ordinary local Lefschetz multiplicity. If all multiplicities

$$\nu_{x'}(g^{-1}f')$$

are equal to 1 at the lifted fixed points, then each local contribution is simply the trace on the corresponding stalk, weighted by the number of lifted fixed points in the Nielsen class.

6. Application to a Lefschetz formula

Let X be proper and let $f : X \to X$ be an endomorphism with isolated fixed points, or more generally with a fixed point scheme satisfying the usual cleanliness hypotheses. Let F be a constructible complex with an endomorphism compatible with f. The Lefschetz formula reads

$$\sum_i (-1)^i \operatorname{Tr}(f^* \mid H^i(X, F)) = \sum_{x \in \operatorname{Fix}(f)} LT_x(f, F).$$

The local term $LT_x(f, F)$ is computed by the Nielsen–Wecken construction after choosing a suitable covering and summing over the lifts.

6.4. Remarks on local terms

If $g \in G$, conjugation by g transports the fixed point data at x' to that at gx', and the local terms transform accordingly. Thus the local contribution depends only on the conjugacy class of the local stabilizer data.

Let x' be a fixed point of a lift and let $G_{x'}$ be its stabilizer. This stabilizer is stable under φ, and the local term may be computed entirely in the complete local ring at x' together with the induced endomorphism of $G_{x'}$. In the tamely ramified case, the Swan contribution vanishes; the local term is then the tame trace multiplied by the geometric fixed point multiplicity.

Lemma 6.4.5. If the ramification at the fixed point is tame and the fixed point multiplicity is simple, the local term is the ordinary trace of the induced endomorphism on the stalk.

When the sheaf F is unramified at all fixed points, the formula simplifies to the standard Lefschetz trace formula with the local terms given by stalk traces and intersection multiplicities.

7. Comments on validity hypotheses

The final comments emphasize that the formula depends on finiteness, properness, and constructibility assumptions. The fixed point locus must be proper over the base, or else cohomology with proper supports must be used. The sheaf must have finite Tor-amplitude in the relevant coefficient category, so that traces are defined. When these hypotheses hold, the local terms are stable under finite coefficient reduction and hence pass to the ℓ-adic limit.

7. Complements: end of Exposé XII

The end of the exposé adds compatibility statements for the local terms. They are invariant under replacing the Galois cover by a dominating cover, additive in distinguished triangles of coefficients, and compatible with the specialization maps used in the proof of the curve case. These facts allow the formula to be applied without making the auxiliary covering part of the final data.

The bibliography of Exposé XII records the topological Nielsen–Wecken sources, the Grothendieck–Verdier trace formalism, and the preceding SGA exposés on local terms and Euler–Poincare formulas.

Exposé XV – Frobenius Morphism and Rationality of L-Functions

Frobenius morphism and rationality of L-functions

1. Frobenius morphism

1.1. Frobenius endomorphism

Let X be a scheme of characteristic $p > 0$. The absolute Frobenius endomorphism

$$F_X : X \to X$$

is the identity on the underlying topological space and raises functions to their pth powers:

$$a \longmapsto a^p.$$

If X is defined over $\mathbb{F}_q$, one also considers the qth Frobenius, obtained by iterating F_X so that functions are sent to a^q.

For a geometric point $\bar{x}$ over a closed point $x \in |X|$, the Frobenius element acts on the geometric stalk of a constructible sheaf. The convention in the trace formula is to use geometric Frobenius, inverse to the arithmetic Frobenius on residue fields.

1.2. Relative Frobenius

Let $X \to S$ be a morphism of schemes in characteristic p. The absolute Frobenius maps of X and S form a commutative square only after twisting the structure morphism. The relative Frobenius is the morphism

$$F_{X/S} : X \to X^{(p)} = X \times_{S,F_S} S.$$

When $S = \operatorname{Spec} k$ for a perfect field k, the Frobenius twist $X^{(p)}$ is another k-scheme, often identified with X after choosing the Frobenius automorphism of k.

The relative Frobenius is functorial in X/S, compatible with products, and finite under the usual finite type hypotheses. It is the geometric morphism which relates point-counting over finite fields to traces on cohomology.

1.3. Effect on a module

If M is a module over a ring A of characteristic p, the Frobenius pullback is

$$F_A^* M = A \otimes_{F_A, A} M.$$

For sheaves this construction is local on X. A semilinear Frobenius structure on a sheaf is an isomorphism between the pullback by Frobenius and the sheaf itself, or dually an action of Frobenius on the stalks. Such structures are the coefficient data in the cohomological interpretation of L-functions.

1.4. Frobenius and cohomology

Let X be separated of finite type over $\mathbb{F}_q$ and let F be a constructible ℓ-adic sheaf. The geometric Frobenius acts on

$$H_c^i(X_{\overline{\mathbb{F}}_q}, F).$$

The Lefschetz trace formula gives

$$\sum_i (-1)^i \operatorname{Tr}\big(F^n \mid H_c^i(X_{\overline{\mathbb{F}}_q}, F)\big) = \sum_{x \in X(\mathbb{F}_{q^n})} \operatorname{Tr}\big(F_x^n \mid F_{\bar{x}}\big).$$

This identity is the bridge between Euler products and determinants of Frobenius on cohomology.

2. Rationality of L-functions

2.1. The theorem

Let X be separated of finite type over $\mathbb{F}_q$ and let F be a constructible $\mathbb{Q}_\ell$-sheaf on X. Define

$$L(X, F; t) = \prod_{x \in |X|} \det(1 - F_x t^{\deg x} \mid F_{\bar{x}})^{-1}.$$

Theorem. The function $L(X, F; t)$ is rational. More precisely,

$$L(X, F; t) = \prod_i \det(1 - Ft \mid H^i_c(X_{\overline{\mathbb{F}}_q}, F))^{(-1)^{i+1}}.$$

This is Grothendieck's cohomological rationality theorem for L-functions. The proof is based on the trace formula and on a reduction which permits one to apply the formula to enough powers of Frobenius.

2.2. A reduction lemma

The reduction lemma says that it suffices to prove the theorem after replacing X by a stratification on which F is locally constant and after replacing F by a sheaf trivialized by a finite cover. Additivity of compactly supported cohomology and multiplicativity of Euler products in distinguished triangles reduce the problem to these cases.

One also reduces to curves in the key step. Given a closed point contribution, one chooses curves passing through the point and uses the Lefschetz formula on those curves. The local computations of Exposés XI and XII control the terms which arise in this passage.

2.3. Proof of the theorem

The logarithm of the Euler product is

$$\log L(X, F; t) = \sum_{n \geq 1} \frac{t^n}{n} \sum_{x \in X(\mathbb{F}_{q^n})} \operatorname{Tr}(F_x^n \mid F_{\bar{x}}).$$

By the Lefschetz trace formula the inner sum is

$$\sum_i (-1)^i \operatorname{Tr}(F^n \mid H_c^i(X_{\overline{\mathbb{F}}_q}, F)).$$

For a finite-dimensional vector space V with endomorphism T one has

$$\exp\left(\sum_{n \geq 1} \operatorname{Tr}(T^n) \frac{t^n}{n}\right) = \det(1 - Tt)^{-1}.$$

Applying this identity to each cohomology group gives the determinant formula, hence rationality.

First reduction

The first reduction is from arbitrary constructible sheaves to sheaves which are locally constant on a dense open subset and zero or controlled on its complement. This is done by noetherian induction on the support and by the localization triangle

$$j_! j^* F \to F \to i_* i^* F \to .$$

Second reduction

The second reduction passes to a finite Galois cover which trivializes the locally constant sheaf. The equivariant form of the trace formula expresses the downstairs trace as an average of traces upstairs, weighted by the relevant group elements. This is where the Nielsen–Wecken and local term formulas enter.

Third reduction

The third reduction handles ramification. By compactifying and using the curve case, local ramification terms are shown to contribute rational factors. The integrality and invariance of the local terms ensure that these factors are independent of auxiliary choices.

End of the proof

After the reductions, the trace formula applies directly. The determinant expression obtained from compactly supported cohomology is a rational function, and the coefficient comparison of logarithmic derivatives identifies it with the Euler product. Thus the L-function is rational.

Bibliography of Exposé XV

The bibliography includes Grothendieck's Bourbaki exposé on the Lefschetz formula and rationality of L-functions, the preceding SGA trace-formula exposés, and the finiteness and constructibility theorems for étalecohomology.

Terminological index

The source closes with a terminological index. In modern English usage the principal entries in this range are: Artin conductor, constructible ℓ-adic sheaf, Euler–Poincare characteristic, Frobenius morphism, geometric Frobenius, Grothendieck group, Lefschetz local term, Nielsen–Wecken class, perfect complex, pseudo-coherent complex, relative Frobenius, Swan representation, and trace formula.

Index of notation

The main notation in this range includes F_X for absolute Frobenius, $F_{X/S}$ for relative Frobenius, H_c^i for compactly supported cohomology, $K(\mathcal{C})$ for a Grothendieck group, Sw_y and Art_y for local Swan and Artin representations, $Z(X, F; t)$ and $L(X, F; t)$ for zeta and L-functions, and $\mathrm{D}^{\mathrm{parf}}$ for perfect complexes.